

Shahrukh Kamal Khan

000998261

BSc. Hons Computer Science

A Comparison of Modelling Methods: Customer Order-In Propensity

In depth research and experimentation of *machine learning models* to *predict probability* of customers choosing “*order in*” option for products at Argos.

Introduction

Artificial intelligence and big data has disruptively changed almost all the industries ranging from telecommunication, health services, medicine and finance due to the need of prediction models making its inclusion almost a necessity in every business (Santis, et al., 2017). With the continuous increase in unstructured and stream data, it is still a challenge for researchers to transform it into meaningful data. McKinsey suggests that companies that use data and business analytics to make decisions are more productive and deliver higher returns on equity than companies that don't (Manyika, et al., 2011).

Argos has a humongous catalogue offering more than 60k products (Argos, 2019), whilst offering 20-30k products in store for collection there is always a need for smart inventory management plan. This serves as a difficult problem due to the number of distinct store fronts of Argos stores, simultaneously also having to *predict* different aspects of numerous products with a customer point of view (Walshaw, 2020). Argos parent company Sainsbury recently started working in with partnership with Accenture to build machine learning solutions on Google Cloud Platform to gain new insights about customers and product trends. They have been able to develop predictive analytics models to spot trends and adjust price range for a better customer experience (Coad, 2019). The intended purpose of this paper is to discuss Argos business problem of calculating customer order in propensity for its products. Different machine learning models and their uses will be discussed in relation to customer order in propensity which is the probability or tendency of a customer to order a product that they desire and similar problems that researchers have worked upon. As in the initial phase of the project, this paper will be mainly focused on supervised machine learning techniques and how they can be used in this project. Starting with a brief description of machine learning and its types, the next paragraph discusses the problem in detail and other similar real world problems faced by organizations.

Problem Description

Machine learning provides automated methods of data analysis which enables us to *predict* future data by spotting patterns in historical data and performing other decision making (Murphy, 2012). Supervised Learning, a sub-category of machine learning, is used to classify data or predict outcome based on labelled historical data accurately. Supervised learning is being used by different organizations to tackle variety of diverse problems ranging from spam filtering, cancer detection to predicting house and stock prices. The category can be halved further into, namely classification and regression. Classification algorithms, as the name suggests, are used to classify data into specific classes by recognizing features or entities to conclude how a class should be defined. Whereas Regression algorithms provide an understanding of dependant and independent variables in a dataset to form up a hypothesis to predict continuous data. Contrary to Supervised Learning, Unsupervised learning is used to analyse previously unknown data to spot trends and group similar

data points in a dataset. The algorithms are used to discover hidden patterns and give insights which can be used to draw conclusion to solve various problems. It is used in exploratory data analysis, customer segmentation, image and object recognition. Clustering, Association Rules and Dimensionality Reduction are some of the common approaches in Unsupervised Learning (IBM Cloud Education, 2020).

With a wide variety of products offered by Argos for collection from stores, it is very difficult to accommodate all the products at every store due to size of stores. Only a few stores have the physical capacity to hold the stock of all the products on sale by the company. However, the customers have an option to “order in” their required product at their nearest store if it is not already available. The physically smallest level of store or the normal stores that we have around us are called spokes which are linked to bigger stores that also act as a support for the spokes to hold stock inside them. These are called hubs. Furthermore the hubs are linked to regional fulfillment centre’s which are linked to distributional centre’s. The ordered products arrive in 24hrs time span from the linked hub if the hub has the product. Not all customers choose to order the products and wait one day instead they buy it from another brand which results as a loss of a customer and possible profit for the company. Argos wants to have an insight to this customer behavior, to predict the probability of customers choosing to order various products enabling the company to know what products to keep at small stores for sale which ideally would be the ones having a lower chance of being ordered whilst having a good demand. Researchers and businesses have worked on similar problems like predicting customer churn, or segmenting their customers to devise custom plans for them.

Businesses use predictive statistical analysis to build propensity models that predict customers who will buy their product, which is similar to our problem at hand. This enables the business to focus on specific customers not all, and to approach them with a customized plan that would yield a higher chance of success rather than approaching every known customer resulting in waste of time and resources, and would not be as fruitful. Predicting customer churn is also one of the problems faced by many organizations i.e. telecommunication companies, retail and banking firms. Business use churn models to create better products, improve services and customer experiences to achieve high customer retention. Classification algorithms like decision forest, logistic regression, neural networks and support vector machines are some of the machine learning techniques commonly used to create churn models (Barga, et al., 2015).

Methodology

The experimentation process follows few basic steps, typically used in data science problems.

1. Defining the business problem.

It is essential to know about the problem in depth, so that it can be further broken down for the ease of solution or to know how to formulate a plan for the solution. The project problem is elaborated in the section above.

2. Data collection and preparation.

Data can be collected through data mining although there is a wide range of sets of data available to public on the internet. Data can be loaded to IDEs in the form of CSV or Excel files. The data is then cleaned to fill up or remove any missing values. Data for this project is collected and prepared by Argos data science team provided to us.

3. Data analysis.

It is important to visualize the data to gain insights and develop an understanding about the data. The knowledge then allows us to decide what model development approach to take. Data can be visualized by creating scatter plots, box plots, histograms and more complex graphs depending on the data and its use case.

4. Feature selection.

Crucial part of data preprocessing is feature/variable selection. The process is to find the right variables for predictive model which is critical in large datasets involving lots of data variables. Using too many variable is useless as it would waste resources and the model would turn out to be extremely specific to the data while overfitting. The model will fail to predict any new data other than the current data set. Different correlation methods like Pearson, Kendall and Spearman are used to find essential features from a dataset.

5. Model Development.

Firstly we have to choose the right algorithm for our problem. One of the tasks of the project is to use and analyze different algorithms to conclude which would be the best approach towards the solution of this problem. Regression and classification algorithms are intended to be researched upon as they fit the problem domain best. The problem can be stated as customer buying a product: yes or no, which suggests the use classification algorithms such as logistic regression, support vector machine, decision trees while calculating the probability of 'yes' and 'no' as well. Similarly the problem can be treated as to predict what number of times a product would be ordered in suggesting the use of regression algorithms predicting continuous output then calculating the probability through the demand of the products and the products actually ordered. Further research is to be done upon this area to determine what algorithms would be best to use as comparison.

6. Model testing and validation.

After the model is trained we have to test it using sample of the dataset that we did not use in training. The sample can vary from 20-30% of the whole data. We select a portion of the data for training and the rest is used for testing. The model is built upon training data then the test feature set is passed to the model for prediction. Finally, the predicted values for the test set are compared with the actual values. This approach gives us a better evaluation on out of sample accuracy however it highly depends on which dataset the model is trained and tested upon.

K-fold Cross Validation is an approach which resolves the issues of dependency and low out of sample accuracy. The data is divided into four folds where in each fold 25% of the data is set as test data and the model is trained on the rest. This continues for the other folds changing the test data, whilst calculating the out of sample accuracy for each fold. Finally, the accuracy is averaged to get a correct evaluation.

7. Model performance.

Evaluation metrics are used to explain the performance of a model. We can compare the predicted values with the actual values to calculate the accuracy of a model. Evaluation metrics provides us an insight to areas that require improvement. Some of the Evaluation

Metrics used for Regression Models are Mean Absolute Error, Mean Squared Error and Root Mean Squared Error. Error is the difference between the data points and the trend line generated by the algorithm.

The project will be based upon python programming language due to its flexibility and ease of use. There are numerous libraries that can be used for different tasks of machine learning from splitting the dataset to training models, some of which are as follows.

NumPy: Math library to work with N-dimensional arrays in python. It is effective and efficient for computation.

SciPy: Collection of numerical algorithms and domain specific toolboxes, including signal processing, optimization, statistics and much more. Used for scientific and high-performance computation.

Matplotlib: Plotting package allowing 2d and 3d plotting.

Pandas: Library providing high-performance and easy to use data structures. Many functions for data importing, manipulation and analysis. It provides data structures and operations for manipulating numerical tables and timeseries.

SciKit Learn (sklearn): Collection of algorithms and tools for machine learning. It has most of the classification, regression and clustering algorithms. Works with NumPy and SciPy. It has great documentation and is easy to implement. Most of the tasks in the machine learning pipeline are already

Evaluation

The intent of this project is to research upon the order in propensity problem by gaining knowledge about various aspects of the data, customer behavior and product behavior in different store types, locations and on different dates which would help to narrow down the list of modelling methods that can serve to be useful. The main objective is to create an application that predicts the probability of customers ordering a product on a given date and location. The main goal can be linked with other explicit objectives which can change during the development cycle of this project, i.e. predicting simple customer order in propensity and collected order in propensity, providing details of comparison in modelling methods on how they work towards the solution of the problem, and concluding the model that is the most beneficial for this scenario. The conclusion will be made with the use of different evaluation metrics and model accuracy explained in the previous, methodology, section.

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